Status	Finished
Started	Tuesday, 20 October 2026, 9:27 PM
Completed	Tuesday, 20 October 2026, 9:31 PM
Duration	3 mins 36 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Deleting elements from a list
b. Sorting elements in a list
c. Inserting elements into a list
d. Determining whether an element is present in a list ✓

Question **2**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O👎
b. O(1)
c. O(log n) ✓
d. O(n log n)

Question **3**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(1)
b. O(log n)
c. O👎 ✓
d. O(n log n)

Question **4**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Binary ✓
- b. Both (1) & (3)
- c. Hash
- d. Linear

Question **5**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By using a hash function
- b. By dividing the list length by two ✓
- c. By comparing each element sequentially
- d. By starting from the first element

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search is more efficient than binary search on large lists.
- b. Linear search requires the list to be sorted.
- c. Linear search divides the list into halves.
- d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **7**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Bubble ✓

b. Insertion

c. Complexity

d. Selection

Question **8**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Sort the list

b. Compare the target element with the middle element in the list ✓

c. Compare the target element with the first element in the list

d. Divide the list into two equal parts

Question **9**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search continues until the end of the list

b. The search backtracks to find duplicate elements

c. The search stops immediately ✓

d. The search starts over from the beginning

Question **10**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 94 and 99

b. 90 and 100

c. 89 and 94

d. 90 and 99 ✓

Question **11**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O👎

b. O(log n)

c. O(n log n)

d. O(1) ✓

Question **12**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Linear search ✓

b. Hash search

c. Binary search

d. Both (1) & (3)

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It is slower than linear search for small lists

b. It requires the list to be sorted ✓

c. It does not work with negative numbers

d. It can only be applied to large lists

Question **14**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and sorted
b. When the list is small and sorted
c. When the list is small and unsorted ✓
d. When the list is large and unsorted

Question **15**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By comparing each element in the list sequentially ✓
b. By dividing the list into halves
c. By using a hash function
d. By sorting the list first